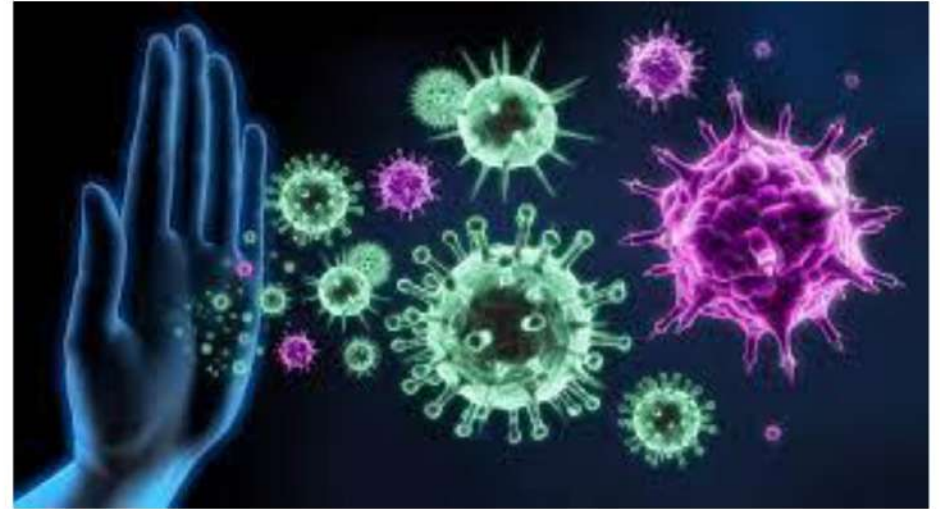


immunology

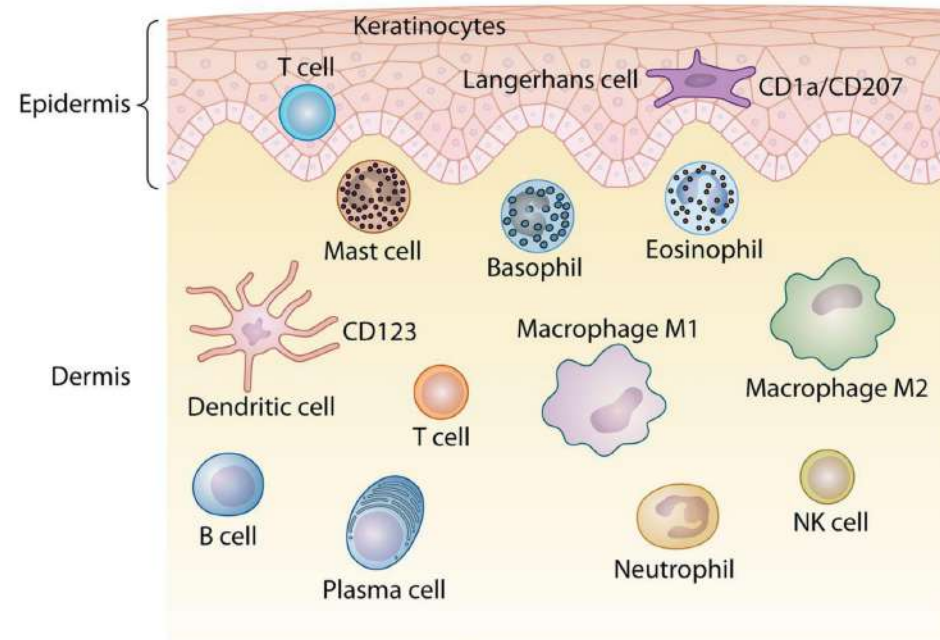
DR EMAN GOMAA

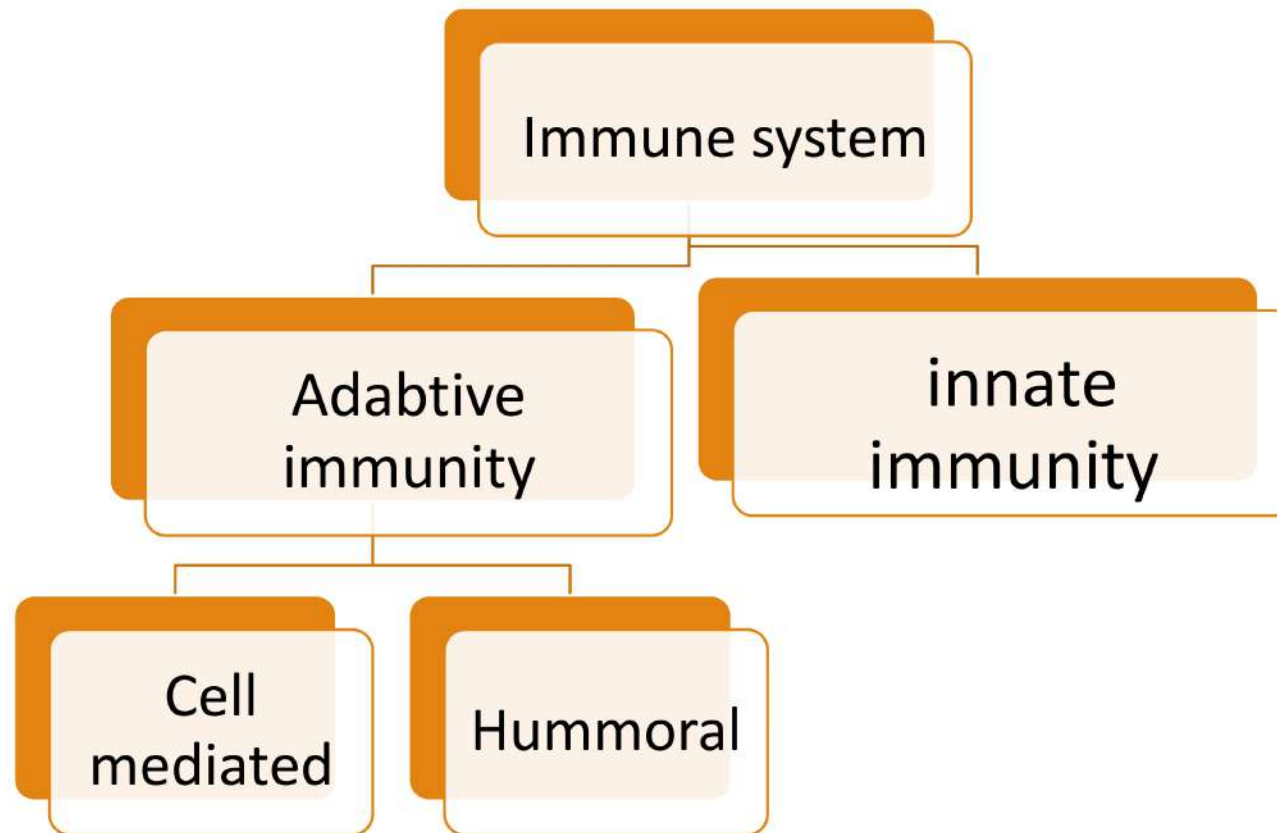
CONSULTANT DERMATOLOGY



What is immunity

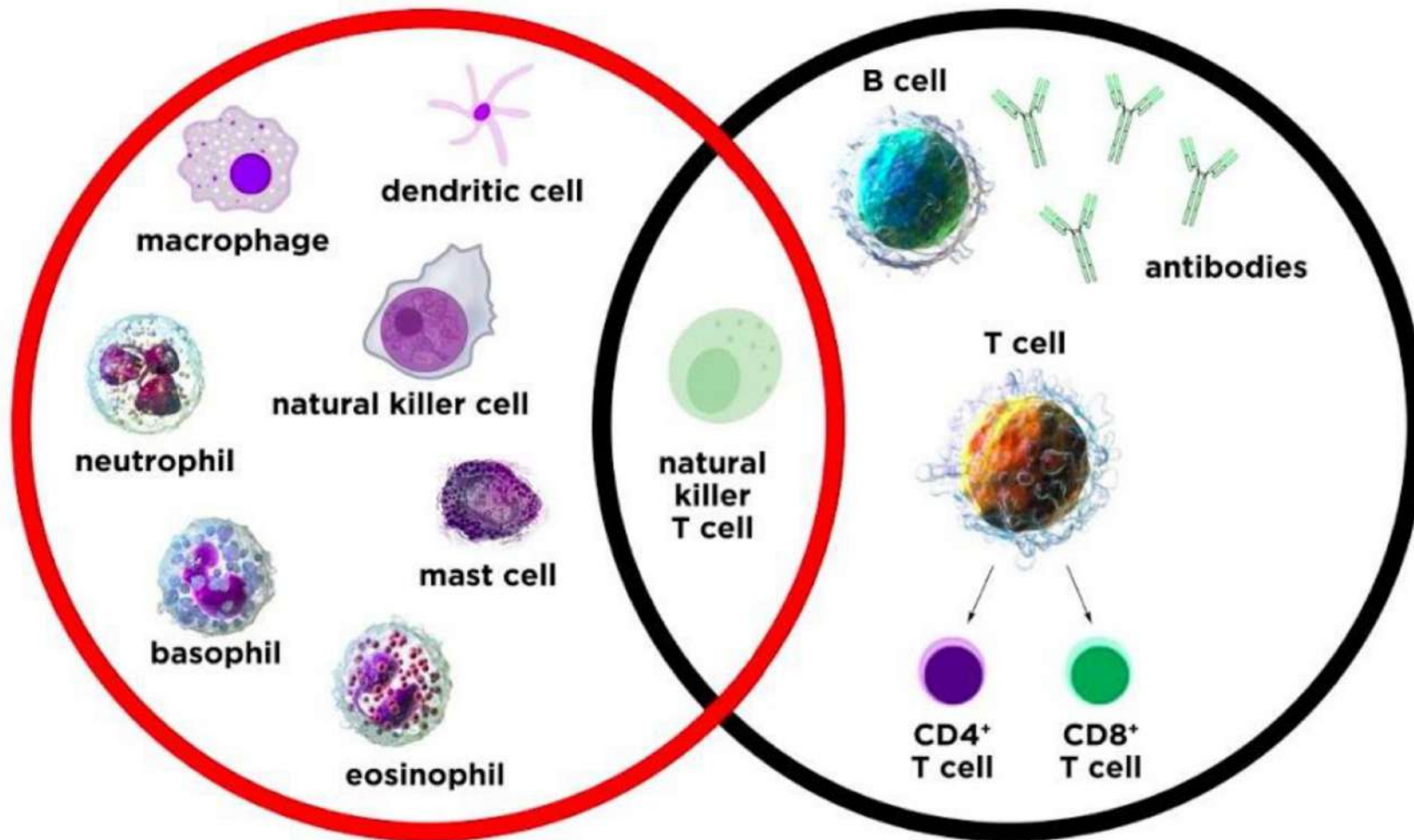
- Immunity is :
all defensive mechanisms that can identify and eliminate foreign threats.





Innate Immunity

Adaptive Immunity

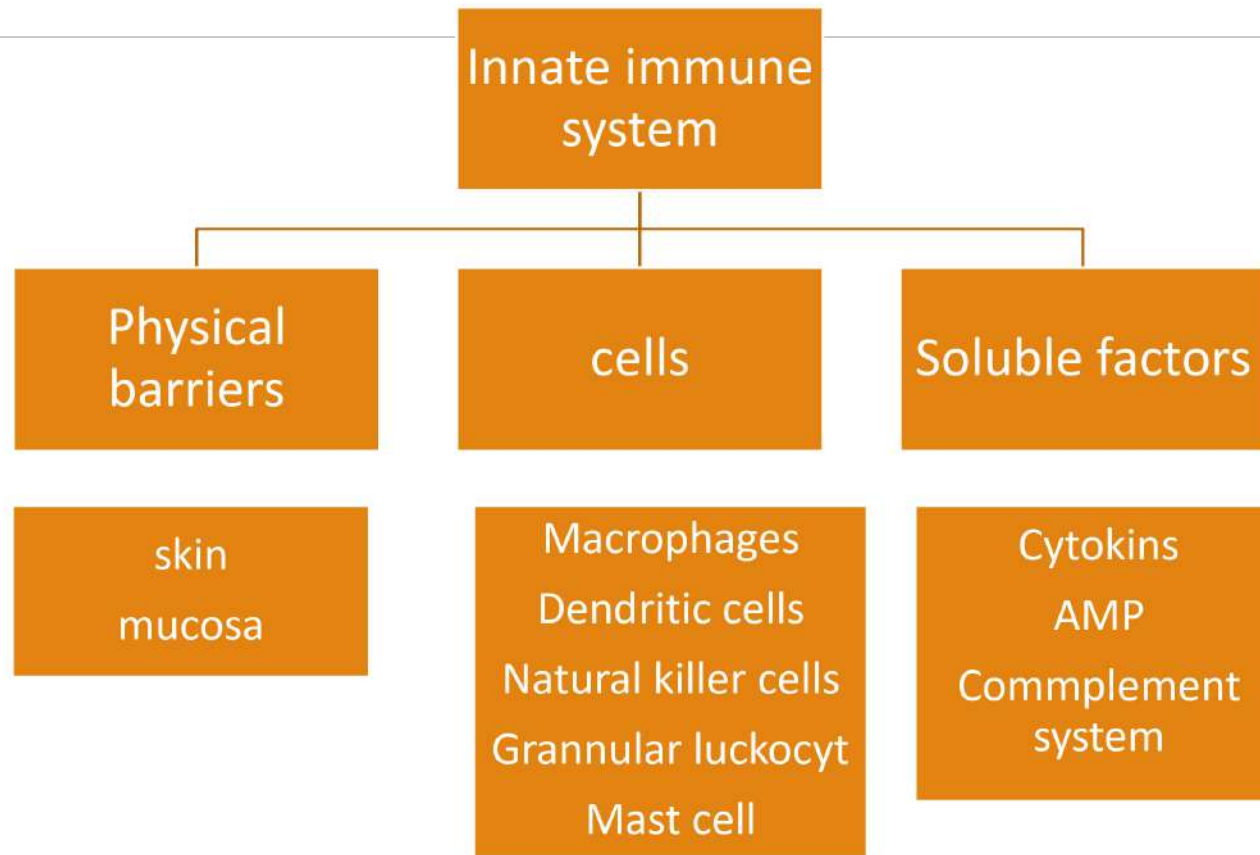


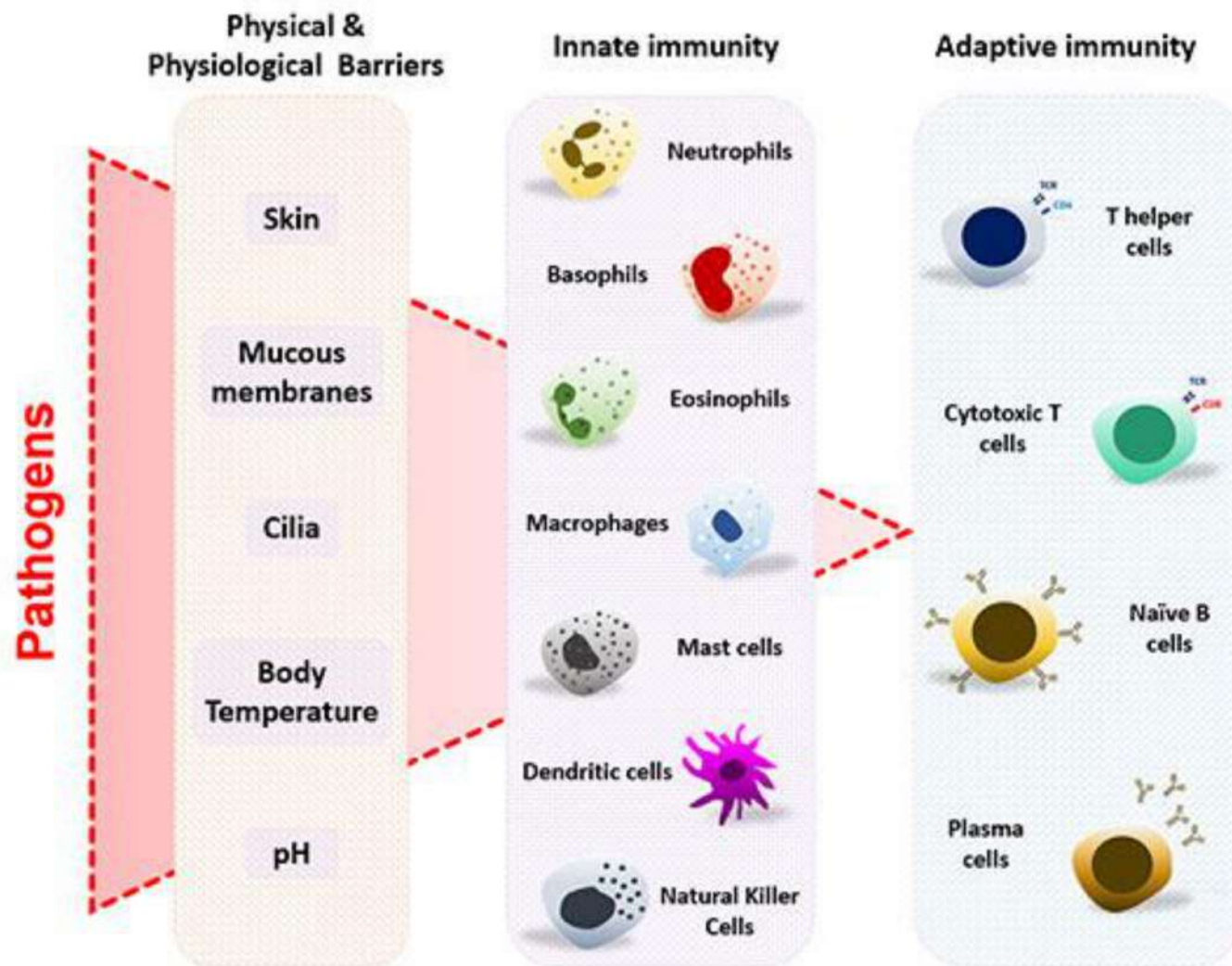
	Innat immunity	Acquired immunity
	1 st line of defense	2 nd line of defense
triggers	Non specific Pathogen associated molecular pattern	Spesific antigen
receptors	PRR TLR	TCR BCR
Action	Min-hours	Days to weeks
memory	No	yes
comunication	Cytokines	chemokines
effect	Ag presentation , complement activation phagocytosis	Cytotoxic cell ,Ab
Ag recognition	Native form	Prcessed form
cells	Macropages , NK ,	B ,Tlymphocytes

Innate immunity

- its inborn with every one . It is 1st line of defense .
- It can differentiate between self and non self antigen and differentiate between pathogenic and non pathogenic .
- Function:
 - Identification and removal of F.B substance
 - Attraction of immune cells to site of infection
 - Activation of complement system .
 - Activation of adaptive immunity through APC .

Componant of innate immunity





Cells of immune system

I. Natural killer cells (NK) :

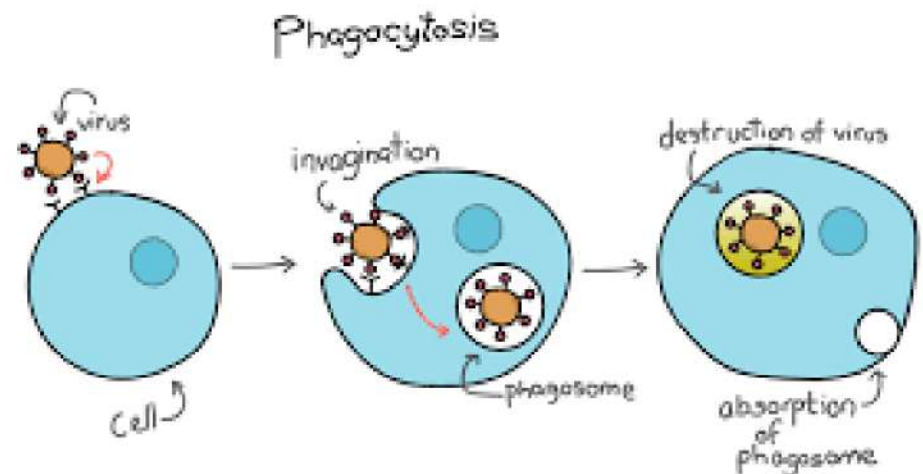
- They are non B ,T lymphocyte ,
- They have non spesefic cytotoxic action
- Kill viraly infected , tumor cells
- Act by two ways
 - Ab depedent cellular cytotoxicity
 - Cytolysis by perforene , granzym



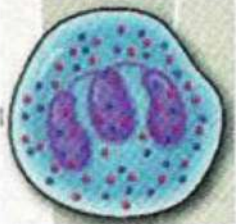
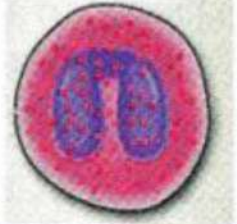
2-Phagocytic cells

- I. **Monocytes** :5% of total leukocytes in blood
migrate to tissue called macrophages

- I. **Macrophages :**
 - I. phagocytosis
 - II. APS
 - III. Secretion of IL1 –IL6
 - IV. Regulation of hemostasis



3- Granulocytes

Neutrophils	Eosinophils	Basophils
		
~ 60% of leukocytes MCQ	0% - 5% of leukocytes	0% - 1% of leukocytes
Both basic and acidic cytoplasmic granules (neutral staining).	Basic (acidophilic / eosinophilic) cytoplasmic granules	Acidic (basophilic) cytoplasmic granules
2-5 nuclear segments i.e. polymorphonuclear (PMN) cells	Bi-lobed nuclei	Bi-lobed nuclei
Neutrophils are phagocytic cells (engulf and destroy bacteria)	Innate and adaptive immune responses to parasitic infections (mainly helminthes). Also, play a role in allergic reactions .	Their granules contain vasoactive amines (e.g. histamine) → smooth muscle contraction and vasodilatation → role in allergic reactions .

- **Mast cells:**

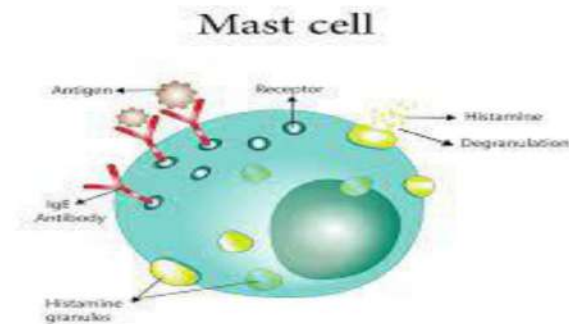
- o Development: derived from precursors in the bone marrow, circulate in the peripheral blood (immature) and finally migrate into tissues,

- o Types:

- A Mucosal mast cells: contain only trypsin (T).

- A Connective tissue mast cells: contain trypsin & chymotrypsin (TC).

- o Functions: immediate-type allergic reactions as urticaria and angioedema



• Pattern recognition receptors

➤ Its a corner stone of innate immunity ,they recognize pathogen associated molecular pattern(pamps).

➤ **Types of PRR:**

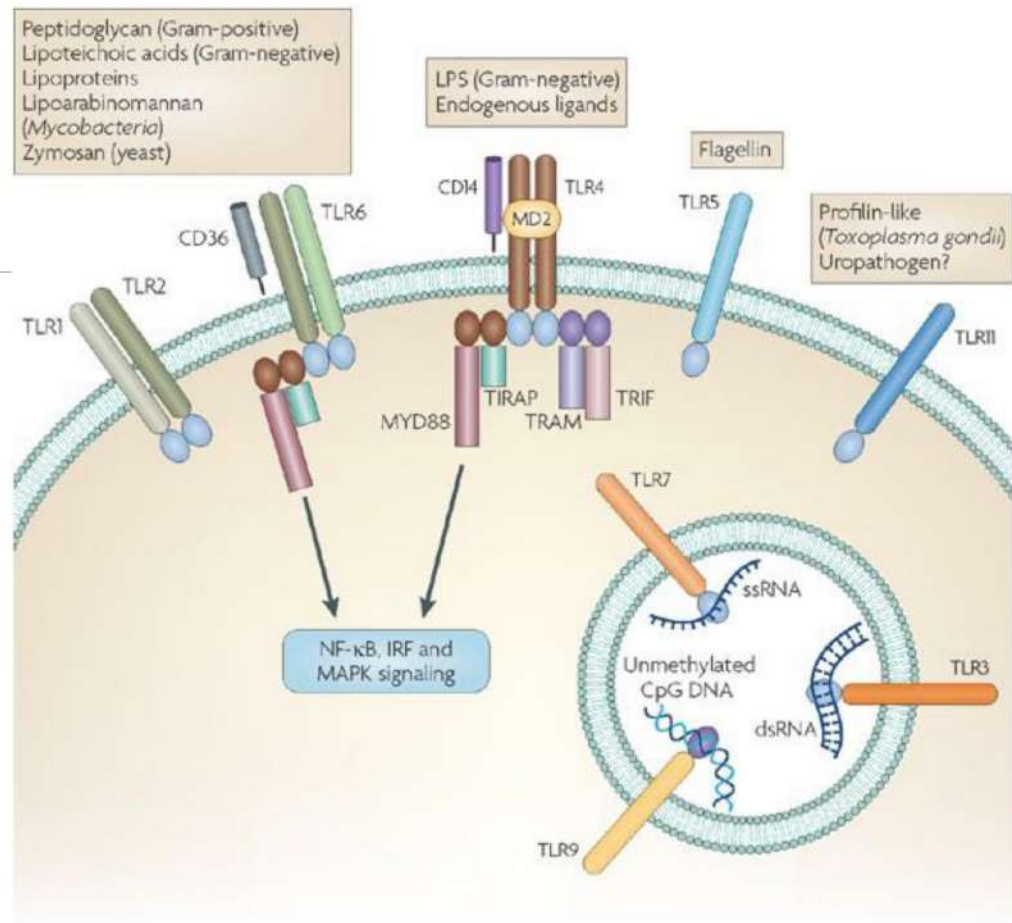
- **TLRS** most important
- Kc mannos binding receptor
- PRP (peptide glycane recognition protein)
- NOD(nucleotide binding oligomerization domain)

- Toll like receptor

- TLRs are a group of glycoproteins surface trans-membrane receptors that help innate immunity to recognize invading microorganisms .
- They are found on immune cells e.g. monocytes/macrophages, DCs, NK cells, granulocytes, B and T cells, and non-immune cells e.g. keratinocytes
- pattern recognition receptors (PRRs). They recognize pathogen-associated molecular patterns (PAMPs). PAMPs are biochemical structures commonly shared by a variety of different pathogens e.g. DNA, RNA, lipopolysaccharides, etc. This makes TLRs non-specific.

At least 10 types of TLRs are found in humans (TLR1 -10). For example:

- o TLR3: recognition of dsRNA (viruses)
- o TLR4: recognition of lipopolysaccharide (Gram-negative bacteria)
- o TLR5: recognition of flagellin (a component of bacterial flagella)

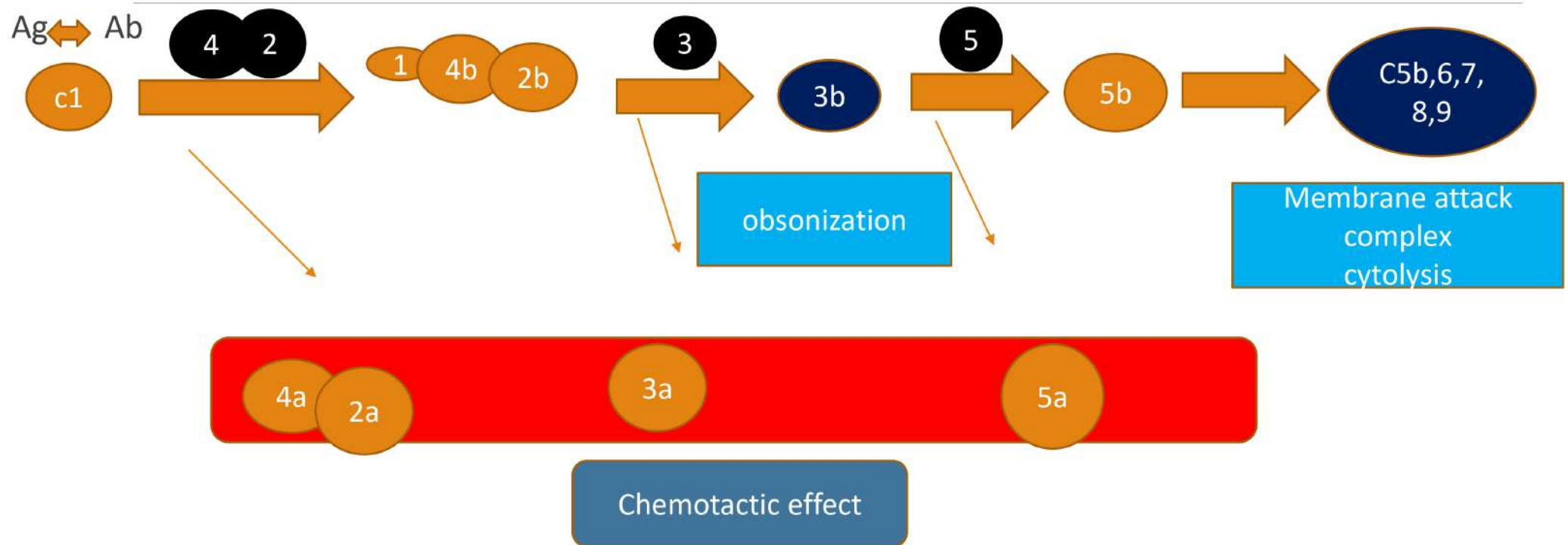


Nature Reviews | Cancer

Complement system

- Def :
biochemical cascade of immune system that help Ab to clear pathogen or mark them for destruction
- Pathways of activation:
 1. Classic pathway :Ag .Ab complex
 2. Alternative pathway
 3. Manan pathway

Complement pathway



Antimicrobial peptide

- group of short peptides , secreted by epithelial surfaces, including the epidermis (KCs, sweat and sebaceous gland)
- Part of innate immunity against organism
- EX : defensins

cathelithidine LL 37

CLINICAL CORRELATION

cytokines

- definition

Substance secreted by many cells , play important roll in intercellular communication .

- types :

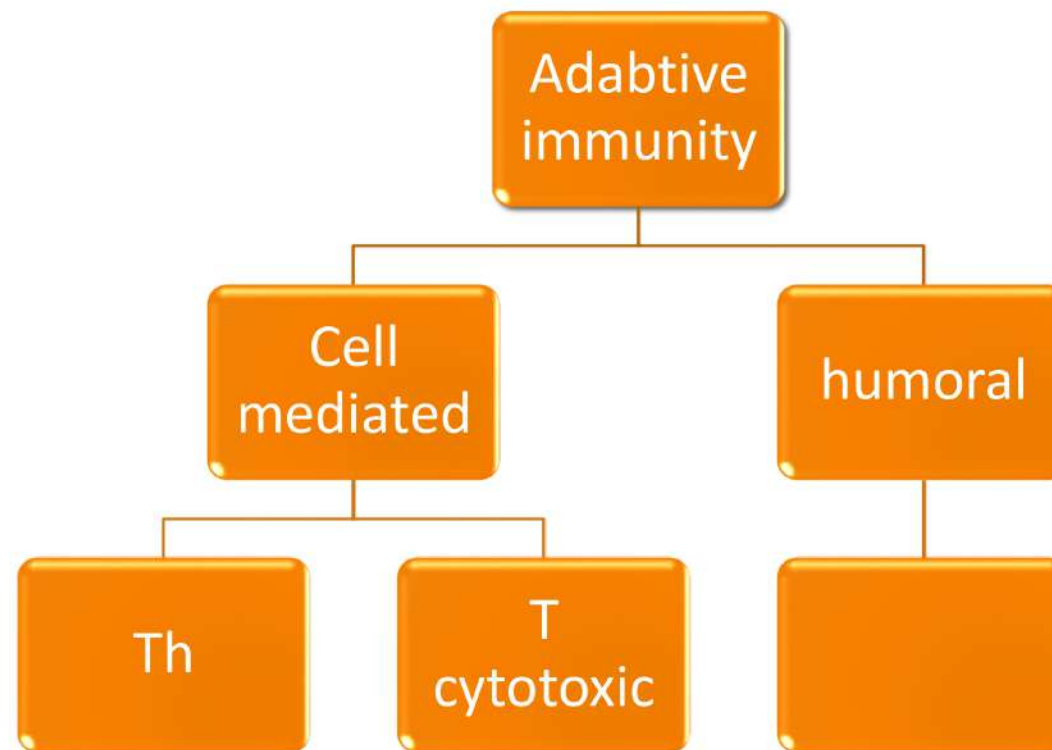
1. Interleukins : produced and exert its action on leukocytes .
2. Interferons: interfere with viral replication .
3. CSF:
4. Chemokines:

Adaptive immunity

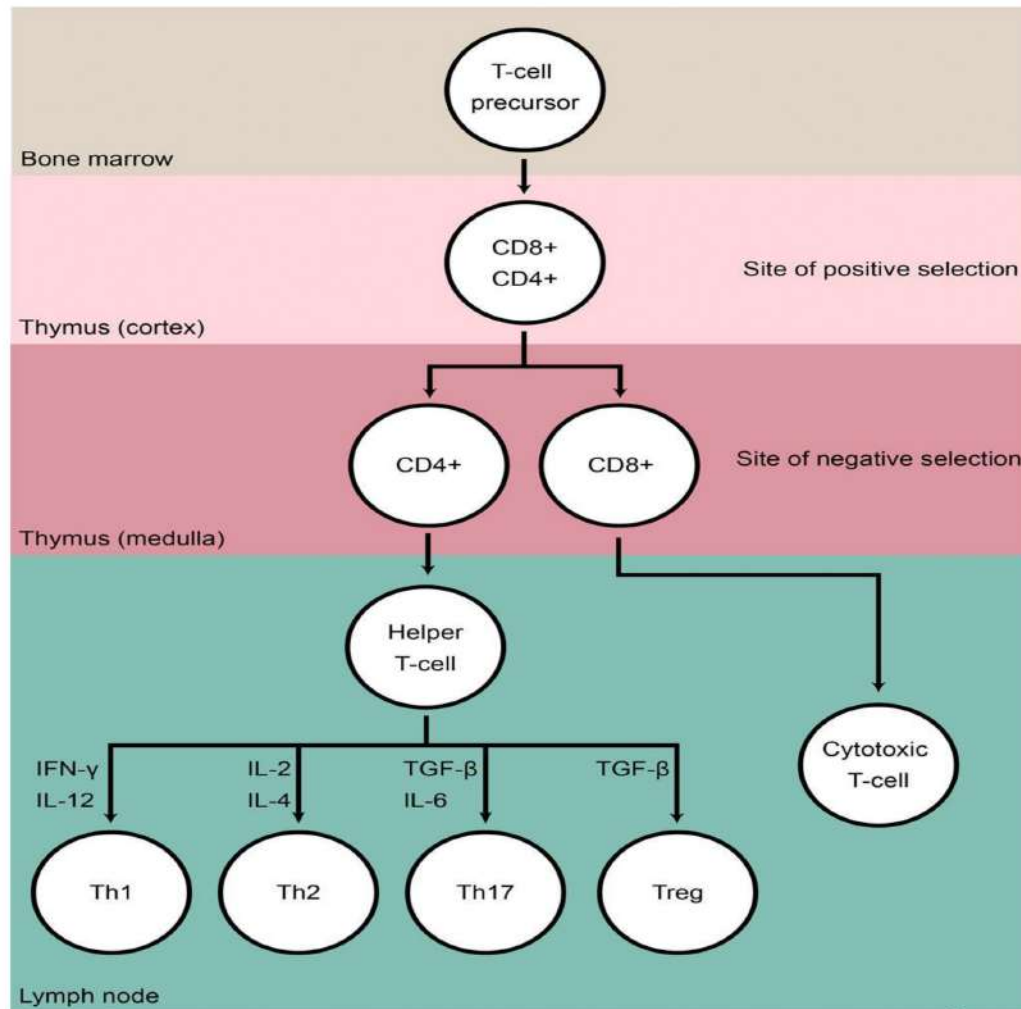
- It is specific long term response .
- It is the 2nd line of defense activated when the antigen overcome innate immunity .
- Mediated by B , T lymphocyte .
- Started by Ag presentation .

Function of adaptive immunity

- Generation of tailored response to eliminate specific pathogens
- Development of immunological memory
- These memory cells called quickly in subsequent infection.



T-Cell Development

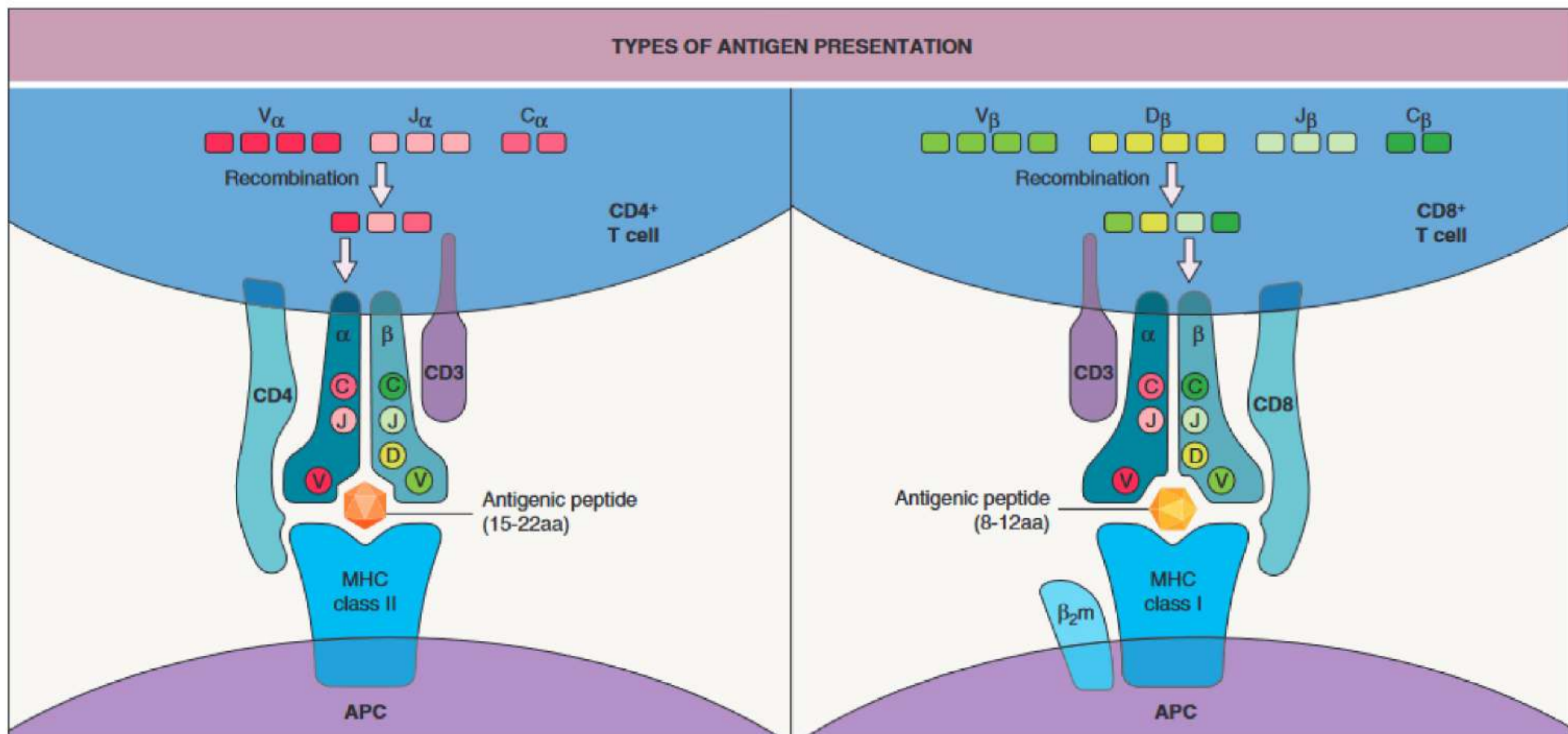


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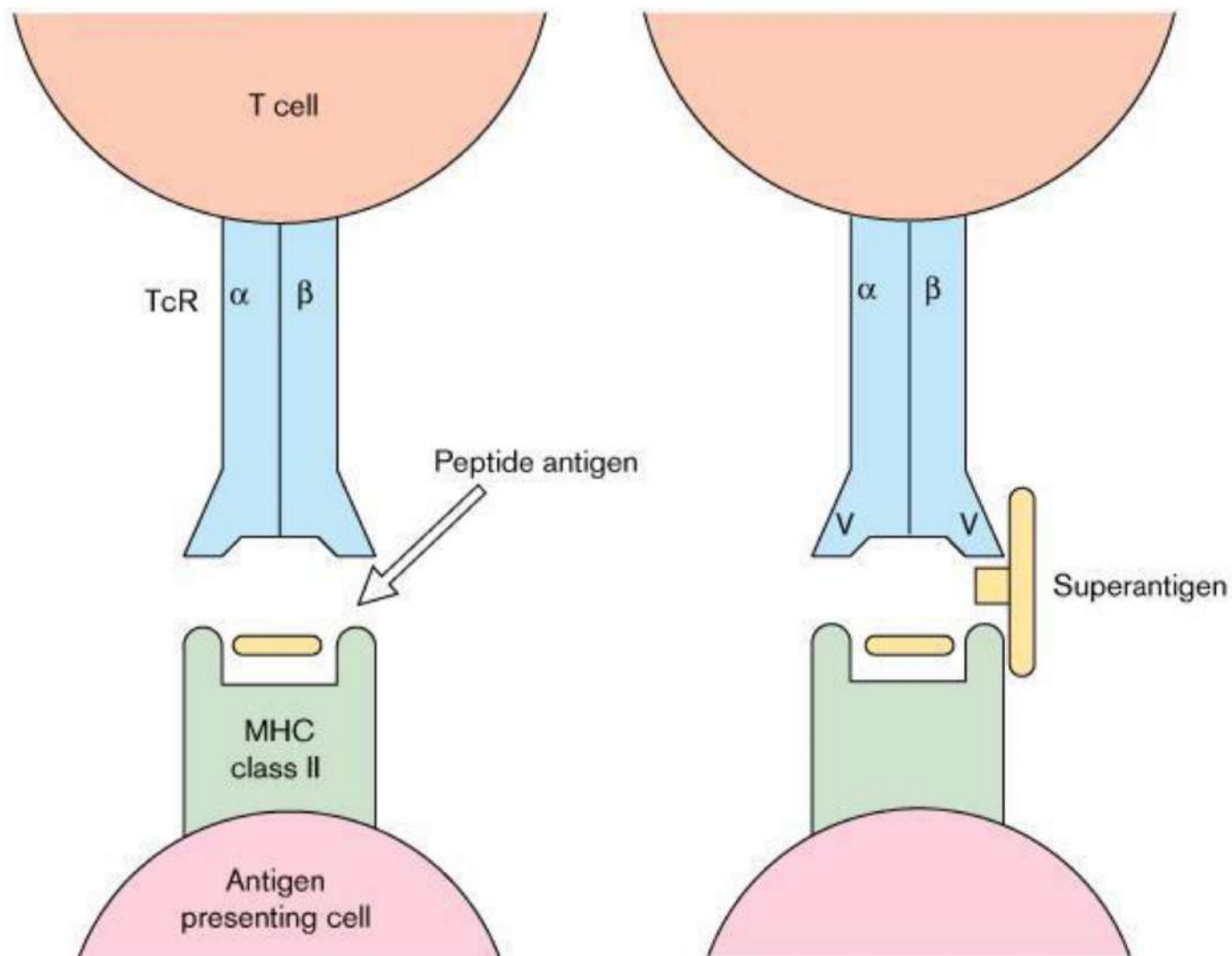
Moises Dominguez

development of T-cells:

T-cell precursors are formed in bone marrow → migrate to the thymus → mature (naive) T cells → secondary lymphoid organs (spleen / paracortical area of lymph nodes).



-cells recognize the antigens through TCRs (α/β chains (mainly) or γ/δ chains (few cells). TCR is associated with CD3



Superantigen :antigen can activate t cell without antigen presentation
EX: toxic shock syndrom

Costimulatory molecules

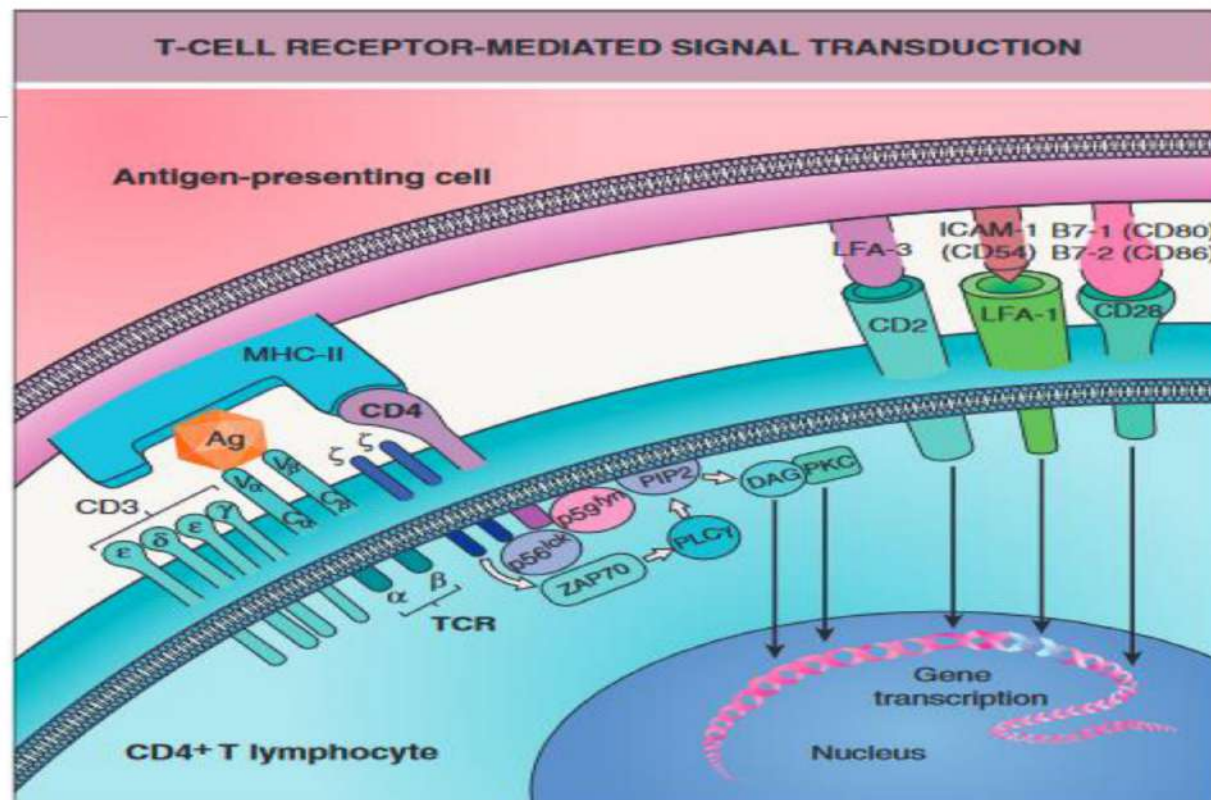
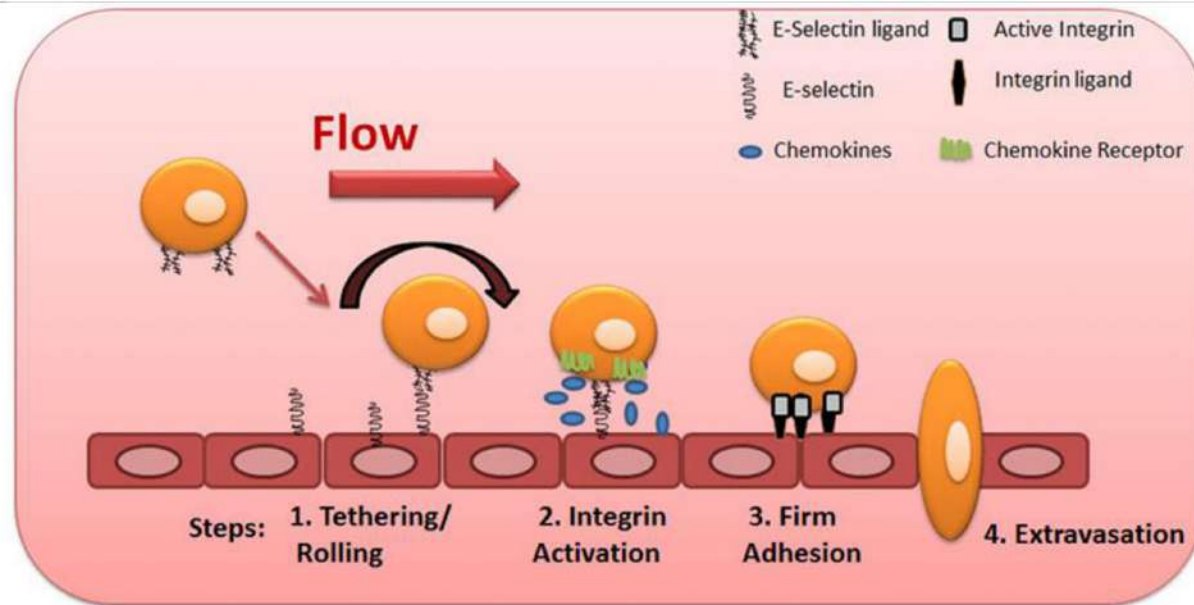


Fig. 4.8 T-cell receptor-mediated signal transduction. Activation of the T-cell receptor (TCR) by presentation of the appropriate antigen by antigen-presenting cells in association with MHC molecules induces a complex signal transduction cascade. The T cell receptor associated signal transduction is

Recruitment (skin-homing) of T-cell



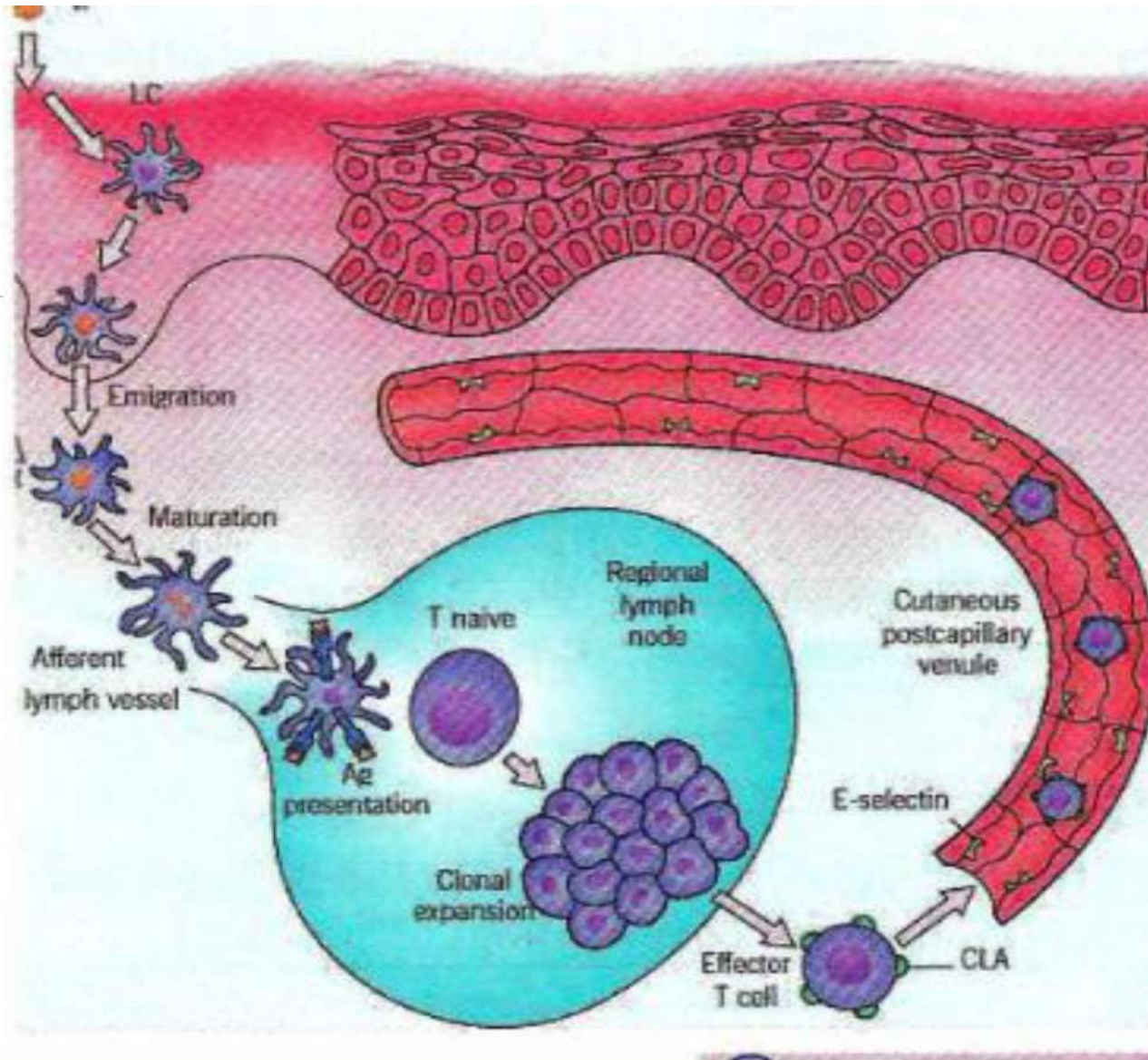
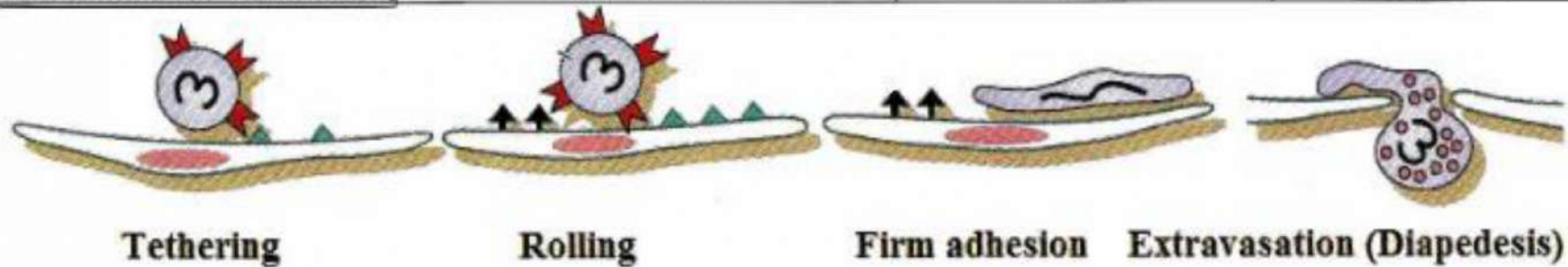


Table 3.2 Adhesion molecules expressed on T-cells and endothelial cells.

T-cell	CLA-1	LFA-1	VLA-4
Endothelial cell	E-selectin	ICAM-1	VCAM-1



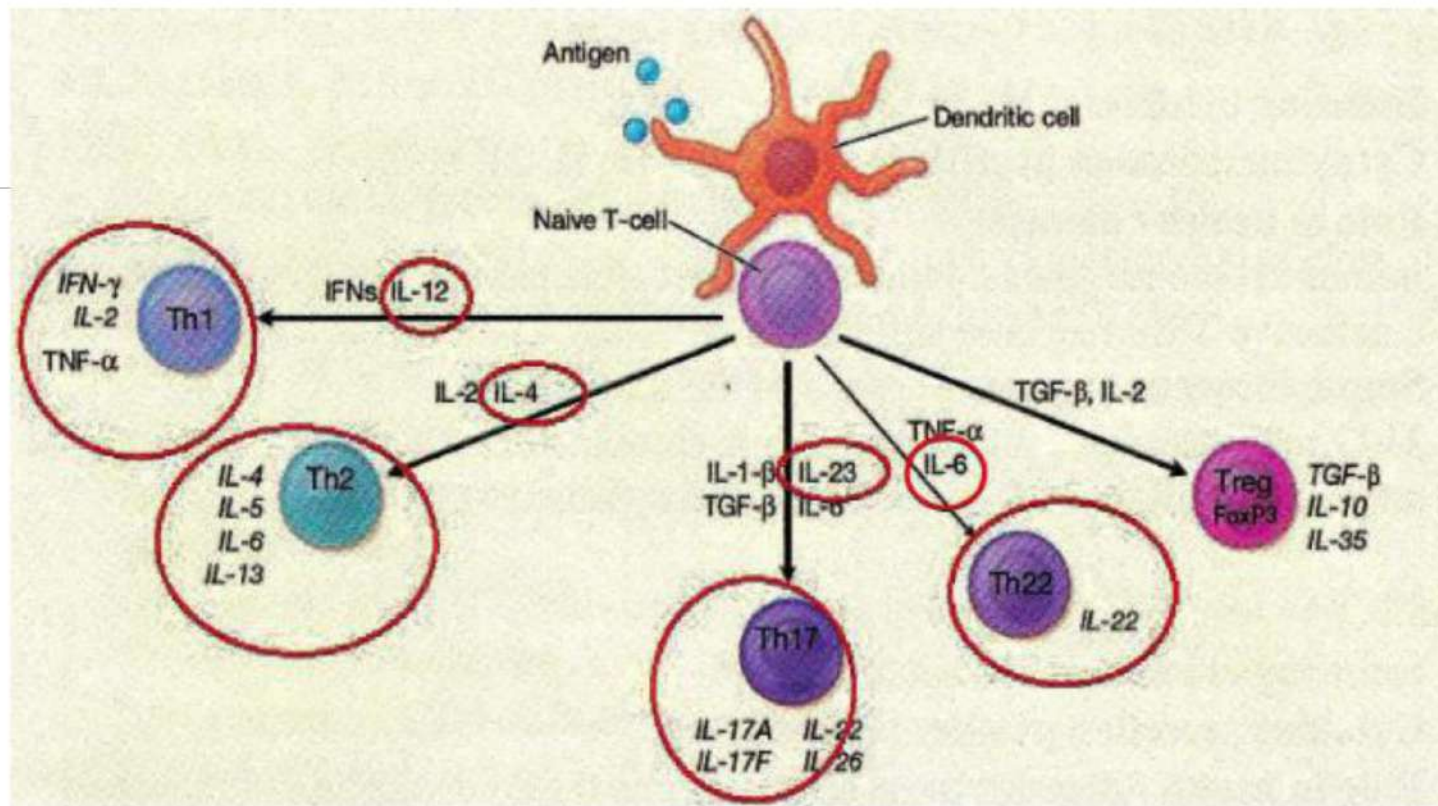
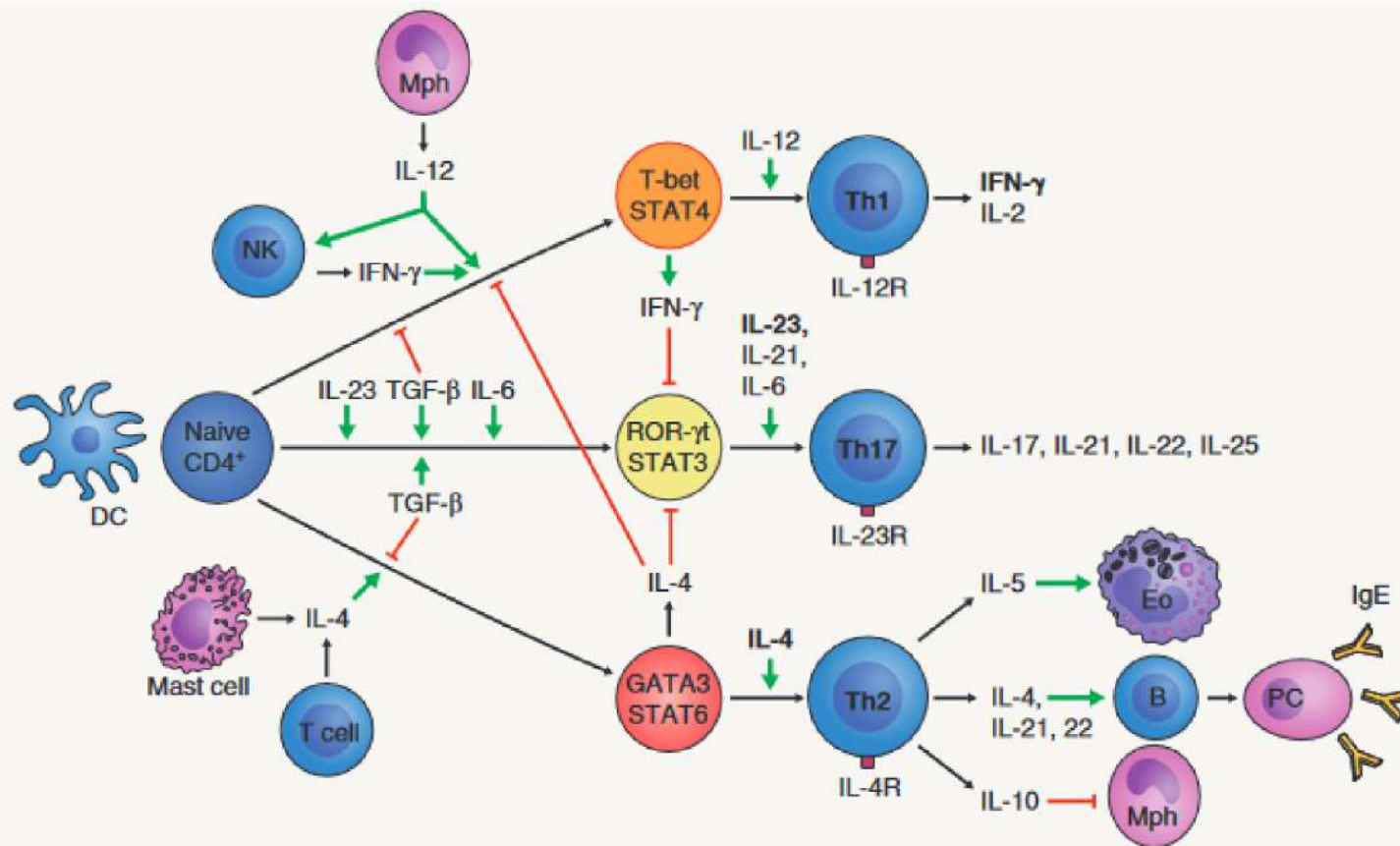
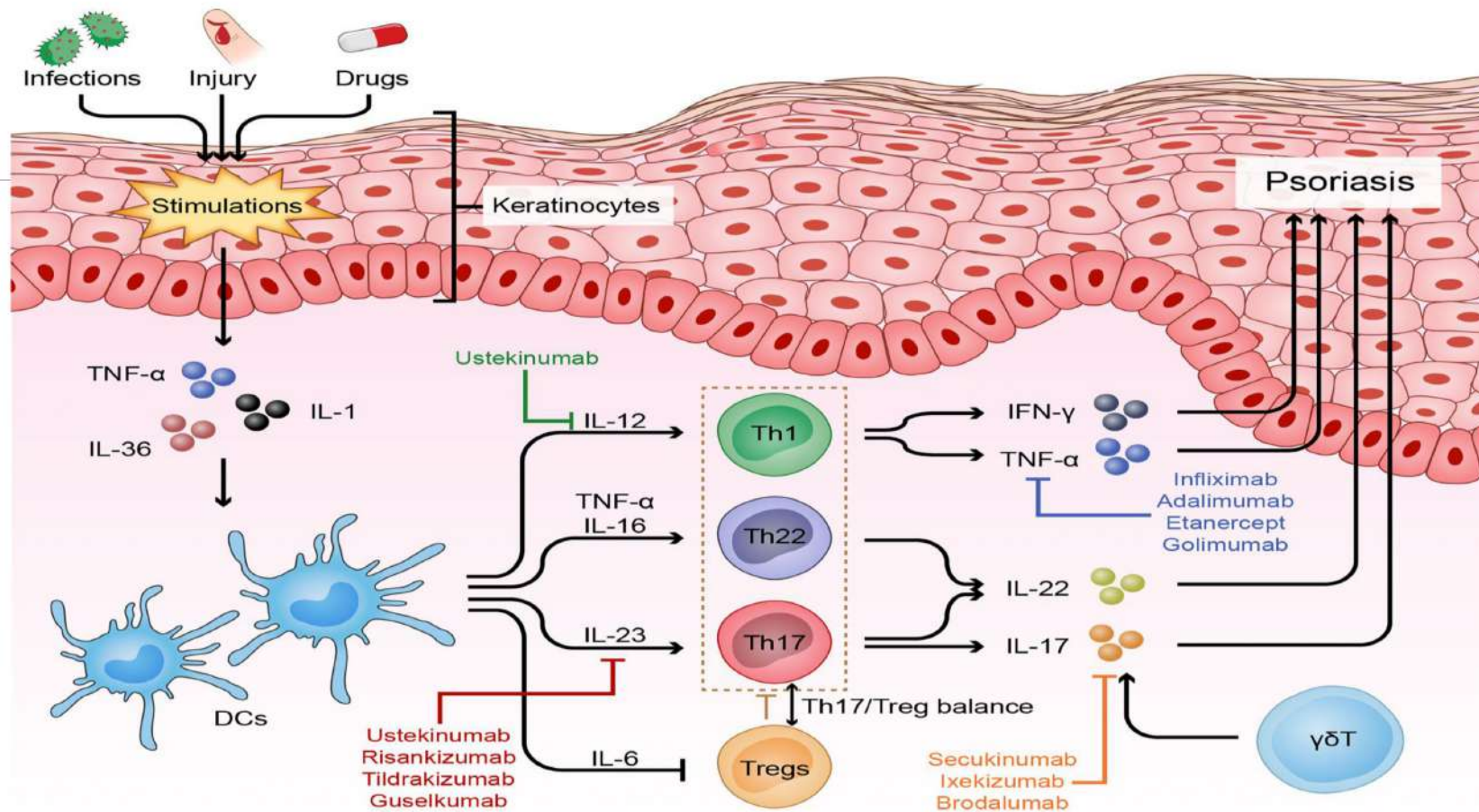


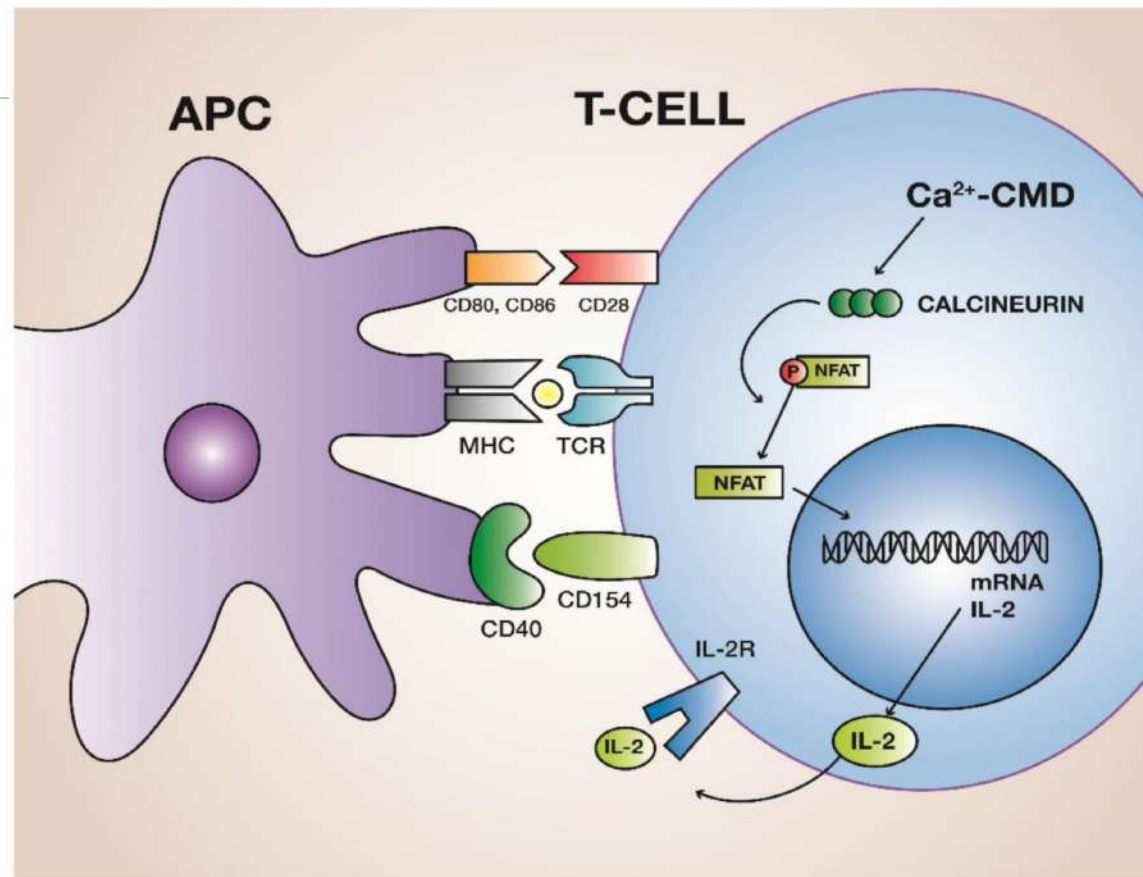
Fig 3.5 Differentiation of naïve Th cells (modified from *Fitzpatrick, 2012*).

DEVELOPMENT OF MURINE TH1, TH2 AND TH17 CELLS



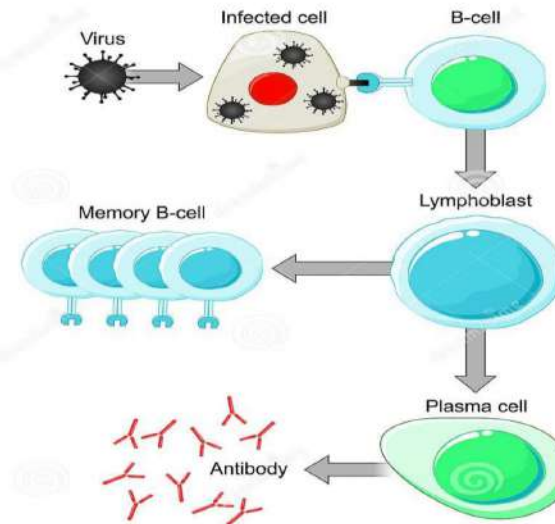


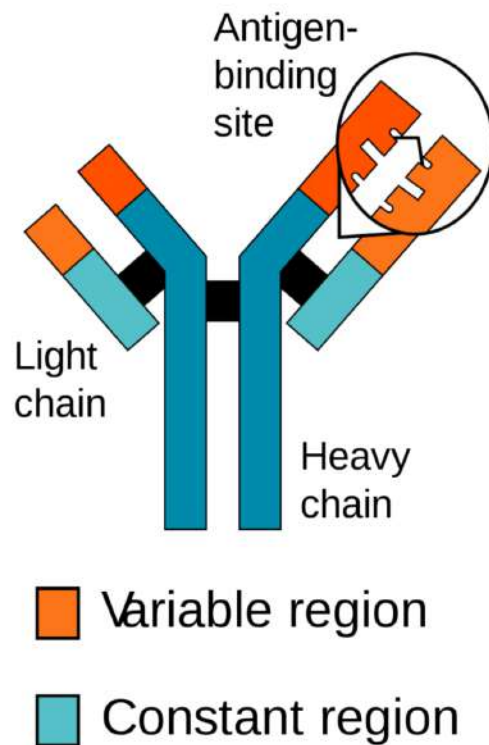
Intra-cellular signaling upon T cell activation



B cell (humoral immunity)

B-cell activation





structure of Igs:

- Two identical heavy chains + two identical light chains.
- Each chain contains a variable region and a constant region.
- There are 5 types of the heavy chains (5 classes of Igs): IgG (Gamma, γ), IgD (Delta, δ), IgE (Epsilon, ϵ), IgA (Alpha, α) and IgM (Mu, μ).
- There are 2 types of light chains: kappa (κ) and lambda (λ).
- Upon proteolysis, each Ig is cleaved into 2 Fab + 1 Fc fragments.

The diagram illustrates the structure of a prototype immunoglobulin molecule, showing its Y-shaped configuration and various regions. The molecule is composed of two heavy chains (orange) and two light chains (yellow), held together by disulfide bonds (S-S). The structure is divided into two main regions: the Fab region (antigen-binding) and the Fc region (constant region).

- Antigen-binding site:** Located at the tips of the Y-shaped arms, where the variable regions of the heavy and light chains interact with an antigen.
- Variable region:** The part of the heavy chain (V_H) and light chain (V_L) that is responsible for antigen specificity.
- Constant region:** The part of the heavy chain (C_H1, C_H2, C_H3) and light chain (C_L) that is responsible for the molecule's biological activity.
- Heavy chain:** The larger orange chain that forms the backbone of the molecule.
- Light chain:** The smaller yellow chain that is associated with the heavy chain.
- N-terminus:** The start of the heavy chain at the top of the Y-arm.
- Hinge region:** A flexible area in the heavy chain that allows the arms to move.
- Complement binding region:** The C_H2 and C_H3 regions of the heavy chain, which are involved in complement activation.
- C-terminus:** The end of the heavy chain at the bottom of the stem.

Types of Igs

IgM: the largest (pentamer) - primary immune response.

- IgG: the most abundant - secondary immune response - mediates most of the autoimmune dermatoses (e.g. pemphigus and pemphigoid).
- IgA: at mucosal surfaces - mediates some bullous autoimmune diseases (IgA pemphigus / linear IgA bullous dermatoses).
- IgE: mediates most immediate allergic and anaphylactic reactions.
- IgD

5 Types of Antibodies

Antibodies or immunoglobulins (Ig) are Y-shaped proteins that recognize unique markers (antigens) on pathogens.



IgA

Secreted into mucous, saliva, tears, colostrum. Tags pathogens for destruction.



IgD

B-cell receptor. Stimulates release of IgM.



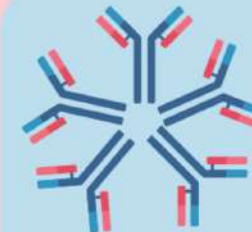
IgE

Binds to mast cells and basophils. Allergy and antiparasitic activity.



IgG

Binds to phagocytes. Main blood antibody for secondary responses. Crosses placenta.



IgM

Fixes complement. Main antibody of primary responses. B-cell receptor. Immune system memory.

Functions of Igs

- Neutralization: preventing antigens from binding to host cell surfaces.
- Agglutination: aggregate formation (limiting antigens mobility, and rendering them more susceptible to destruction).
- Complement activation: classical pathway
- Opsonization: enhancement of phagocytosis.
- Antibody-dependent cell-mediated cytotoxicity (ADCC):

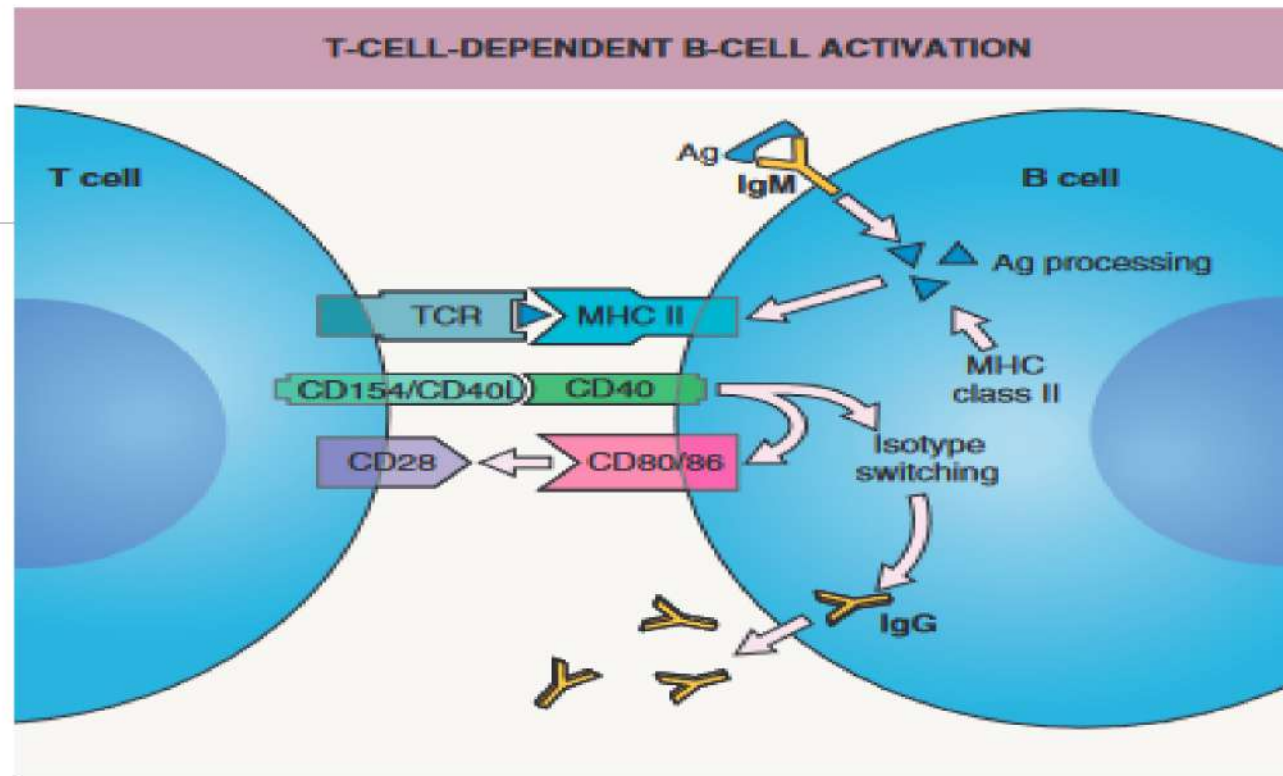


Fig. 4.13 T-cell-dependent B-cell activation. B cells present the antigen in association with MHC II molecules to CD4⁺T cells. This results in the upregulation of CD154/CD40 ligand which interacts with CD40 expressed on B cells. CD40–CD40 ligand interaction causes isotype switching of immunoglobulins and upregulation of CD80/86 which interacts with CD28 expressed on T cells. This results in further activation of T cells. Ag, antigen; CD40L, CD40 ligand; Ig, immunoglobulin; MHC II, major histocompatibility complex II.

3- The pharmacologic activity of tacrolimus includes:

- A. Phosphorylation of NFAT (nuclear factor of transcription)
- B. Binding and inhibition of NF kappa B

-
- C. Inhibition of interleukin-1 gene transcription
 - D. Activation of calcineurin
 - E. Inhibition of interleukin-2 gene transcription

►E

Tacrolimus is a macrolide similar to cyclosporine and pimecrolimus (Elidel). Calcineurin is a phosphatase that dephosphorylates subunit on NFAT (nuclear factor of activated T cells). Dephosphorylated NFAT is active and causes the increase in IL2. Tacrolimus binds macrophillin1 which blocks calcineurin so NFAT is not dephosphorylated ie it remains phosphorylated and is inactive therefore preventing transcription of IL 2.

2- A 24 year old female patient is referred for management of chronic idiopathic urticaria. Many cases of this disease are associated with autoantibodies against what?

- A. IgE receptor
- B. Tryptase
- C. Histamine receptor
- D. TNF-alpha receptor
- E. IL-6 receptor

1 -The immunoglobulin most commonly found in mucous secretions is:

- A. IgA
- B. IgD
- C. IgE
- D. IgG
- E. IgM

►A

IgA is found in mucous membrane secretions and is able to agglutinate antigens and activate the alternate but not the classic complement pathway. IgG is the antibody that can cross the placenta and the most common antibody found in circulation. IgD is not found in circulation other than in hyper-IgD syndrome, an autosomal recessive disorder caused by mutations in the mevalonate kinase gene. A significant elevation of serum IgD is seen in 95% of these patients. IgE is an anaphylactic antibody that is involved in nearly all immediate allergic and anaphylactic type reactions. IgM is the antibody produced in the early stages of antibody responses. It is a pentamer which can agglutinate antigen and activate the classic complement pathway.

11- IL-23 plays a critical role in the pathogenesis of psoriasis. Which of the following cytokines is critical for IL-23-mediated epidermal hyperplasia in psoriasis?

- A. IL-2
- B. IL-12
- C. IL-17A
- D. TNF-alpha
- E. IFN-gamma

Correct choice: C. IL-17A

Explanation: IL-23 and Th17 cells producing IL-17A and IL-22 are found in excess in skin affected by psoriasis. IL-6, IL-22, and IL-17A have all been shown to be critical in mediating epidermal hyperplasia in psoriasis in response to IL-23.

**THANK
YOU**

